

02AMO2

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Problem

Let ABC be a triangle such that

$$\left(\cot \frac{A}{2}\right)^2 + \left(2 \cot \frac{B}{2}\right)^2 + \left(3 \cot \frac{C}{2}\right)^2 = \left(\frac{6s}{7r}\right)^2,$$

where s and r denote its semiperimeter and its inradius, respectively. Prove that triangle ABC is similar to a triangle T whose side lengths are all positive integers with no common divisors and determine these integers.

Solution. Recall the identity $\cot \frac{A}{2} = \frac{s-a}{r}$. Substituting this into the given equation:

$$\left(\frac{s-a}{r}\right)^2 + 4\left(\frac{s-b}{r}\right)^2 + 9\left(\frac{s-c}{r}\right)^2 = \left(\frac{6s}{7r}\right)^2.$$

Let $x = s - a$, $y = s - b$, and $z = s - c$. Then $x + y + z = s$. The equation simplifies to:

$$z^2 + 4x^2 + 9y^2 = \frac{36}{49}(x + y + z)^2.$$

Applying the Cauchy-Schwarz inequality:

$$(z^2 + 4x^2 + 9y^2) \left(1 + \frac{1}{4} + \frac{1}{9}\right) \geq (z + x + y)^2.$$

Since $1 + \frac{1}{4} + \frac{1}{9} = \frac{49}{36}$, we have:

$$z^2 + 4x^2 + 9y^2 \geq \frac{36}{49}(x + y + z)^2.$$

Equality holds when $z/1 = 2x/(1/2) = 3y/(1/3)$, or $z = 4x = 9y$. Letting $z = t$, we have $x = t/4$ and $y = t/9$. Then $s = t/4 + t/9 + t = \frac{49}{36}t$. The sides are: $a = s - x = \frac{49}{36}t - \frac{9}{36}t = \frac{40}{36}t = \frac{10}{9}t$, $b = s - y = \frac{49}{36}t - \frac{4}{36}t = \frac{45}{36}t = \frac{5}{4}t$, $c = s - z = \frac{49}{36}t - \frac{36}{36}t = \frac{13}{36}t$. Multiplying by $36/t$, we obtain the integers 40, 45, 13.